

# Workshop 8

Friday, 12:00pm

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In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

**Specific leaf area** is a continuous variable measured in  $\text{cm}^2/\text{g}$ .

**Water treatment** is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

**Species** is a categorical variable (again, a *factor*) with 6 levels.

| Species name                      | Species code | Common name        |
|-----------------------------------|--------------|--------------------|
| <i>Encelia californica</i>        | ENCCAL       | Bush sunflower     |
| <i>Eschscholzia californica</i>   | ESCCAL       | California poppy   |
| <i>Penstemon centranthifolius</i> | PENCEN       | Scarlet bugler     |
| <i>Grindelia camporum</i>         | GRINCAM      | Gumweed            |
| <i>Salvia leucophylla</i>         | SALLEU       | Purple sage        |
| <i>Stipa pulchra</i>              | STIPUL       | Purple needlegrass |
| <i>Lotus scoparius</i>            | LOTSCO       | Deerweed           |

## 1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMin) # model selection

# read in your data here

drought_exp <- read_xlsx(
  here("data", "Valliere_etal_EcoApps_Data.xlsx"),
  sheet = "First Harvest"
)
```

```
# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
salleu_col <- "#6D397D"
enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
```

```

escscal_col <- "#515481"
gricam_col <- "#6C91BD"

# storing colors to use for water treatments
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw() +
            theme(text = element_text(size = 12)))

```

## 2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```

# create new object for clean data
drought_exp_clean <- drought_exp |>
  # lowercase and underscores in column names
  clean_names() |>
  # creating new column with full scientific names
  # species codes to species names
  mutate(species_name = recode_values(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
    "STIPUL" ~ "Stipa pulchra", # purple needlegrass
    "LOTSCO" ~ "Lotus scoparius" # deerweed
  )) |>
  # creating new column with full treatment names
  # water codes to water treatment
  mutate(water_treatment = recode_values(
    water,
    "WW" ~ "Well watered",
    "DS" ~ "Drought stressed"
  )) |>
  # reordering levels of categorical variable (species name)

```

```
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                   "Lotus scoparius",
                                   "Penstemon centranthifolius",
                                   "Salvia leucophylla",
                                   "Encelia californica",
                                   "Stipa pulchra",
                                   "Eschscholzia californica",
                                   "Grindelia camporum")) |>

# reordering water treatment levels
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                      "Drought stressed",
                                      "Well watered")) |>

# selecting specific columns for the data frame
select(species_name, water_treatment, sla, total_g)
```

Insert code to double check the structure of the data frame here.

```
#| label: checking-structure

str(drought_exp_clean)
```

```
tibble [70 x 4] (S3: tbl_df/tbl/data.frame)
 $ species_name   : Factor w/ 7 levels "Lotus scoparius",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ water_treatment: Factor w/ 2 levels "Drought stressed",...: 2 2 2 2 2 1 1 1 1 1 ...
 $ sla            : num [1:70] 170 215 209 216 222 ...
 $ total_g        : num [1:70] 0.455 0.329 0.45 0.359 0.352 ...
```

Insert code to display 10 random rows from the data frame here.

```
slice_sample(
  drought_exp_clean, # using clean data fram
  n = 10 # show 10 row
)
```

```
# A tibble: 10 x 4
  species_name      water_treatment    sla total_g
  <fct>            <fct>          <dbl>  <dbl>
1 Lotus scoparius  Well watered    121.   0.0887
```

|    |                            |                  |      |        |
|----|----------------------------|------------------|------|--------|
| 2  | Lotus scoparius            | Well watered     | 123. | 0.182  |
| 3  | Eschscholzia californica   | Drought stressed | 107. | 0.302  |
| 4  | Lotus scoparius            | Drought stressed | 90.9 | 0.0796 |
| 5  | Stipa pulchra              | Drought stressed | 135. | 0.332  |
| 6  | Encelia californica        | Drought stressed | 171. | 0.259  |
| 7  | Stipa pulchra              | Drought stressed | 176. | 0.181  |
| 8  | Stipa pulchra              | Well watered     | 161. | 0.376  |
| 9  | Penstemon centranthifolius | Drought stressed | 143. | 0.0881 |
| 10 | Grindelia camporum         | Well watered     | 213. | 0.678  |

### 3. Visualizing data

#### What is the relationship between SLA and plant mass?

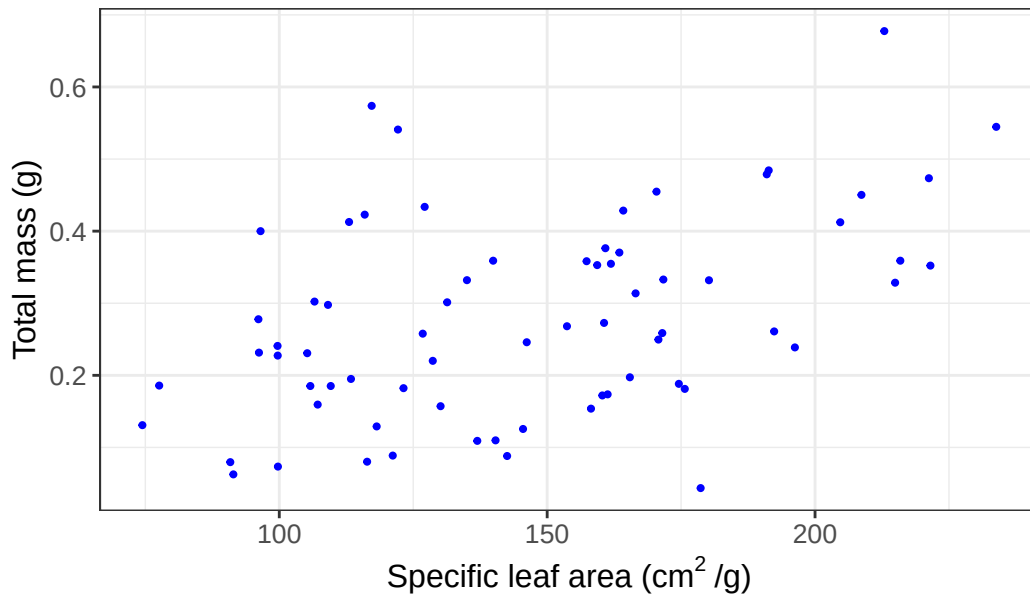
Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass \_\_\_\_\_).

```
#base layer: ggplot
ggplot(data = drought_exp_clean,
       mapping = aes(
         x = sla, # x-axis: specific leaf area
         y = total_g, # y-axis: total plant mass
       )) +
  # scatter plot
  geom_point(
    color = "blue",
    size = 0.75
  ) +
  # title and axes titles
  labs(
    title = "As SLA increases, total mass increases",
    x = expression("Specific leaf area (cm"^-2~/g)"),
    y = "Total mass (g)"
  )
```

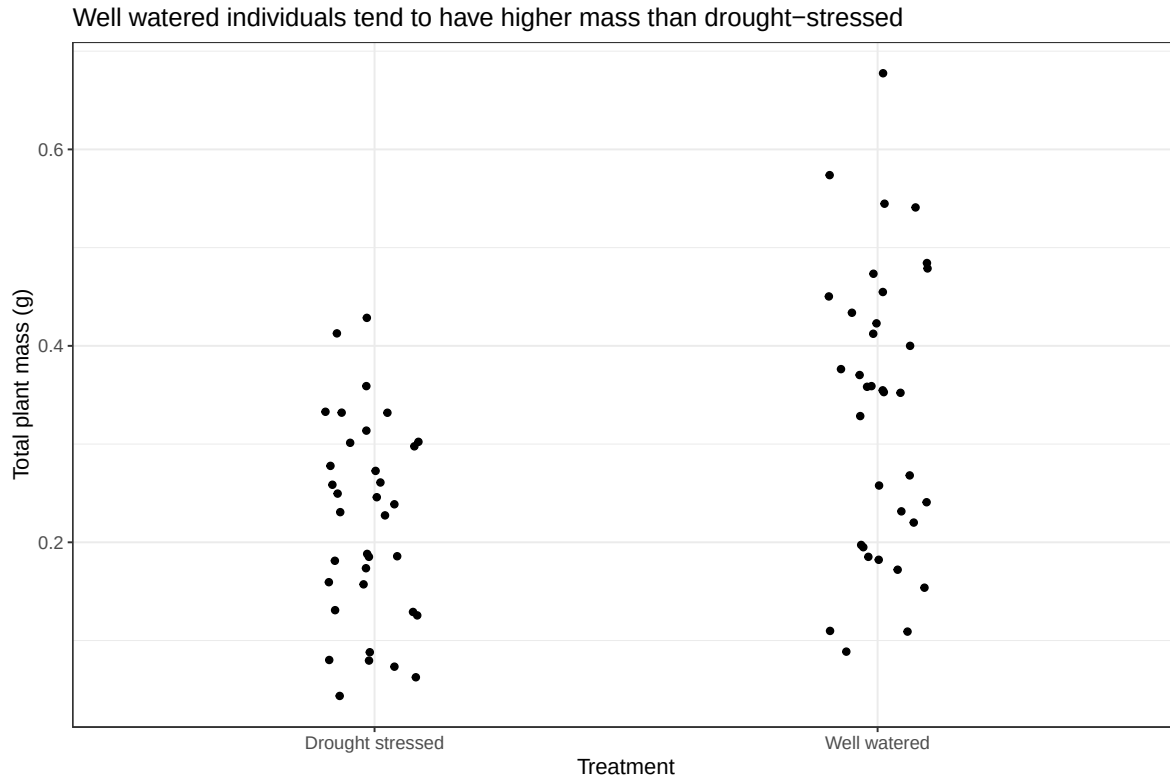
As SLA increases, total mass increases



**What are the differences in total mass between water treatments?**

Make a jitterplot to visualize the differences in plant mass between water treatments.

```
ggplot(data = drought_exp_clean,
       mapping = aes(
         x = water_treatment,
         y = total_g,
       )) +
  geom_jitter(
    height = 0,
    width = 0.1,
  ) +
  labs(
    title = "Well watered individuals tend to have higher mass than drought-stressed",
    x = expression("Treatment"),
    y = "Total plant mass (g)"
  ) +
  theme(legend.position = "none")
```



Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between water treatments.

### What are the differences in total mass between species?

Make a jitterplot to visualize the differences in plant mass between species.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

Bonus: order the x-axis by decreasing mean plant mass.

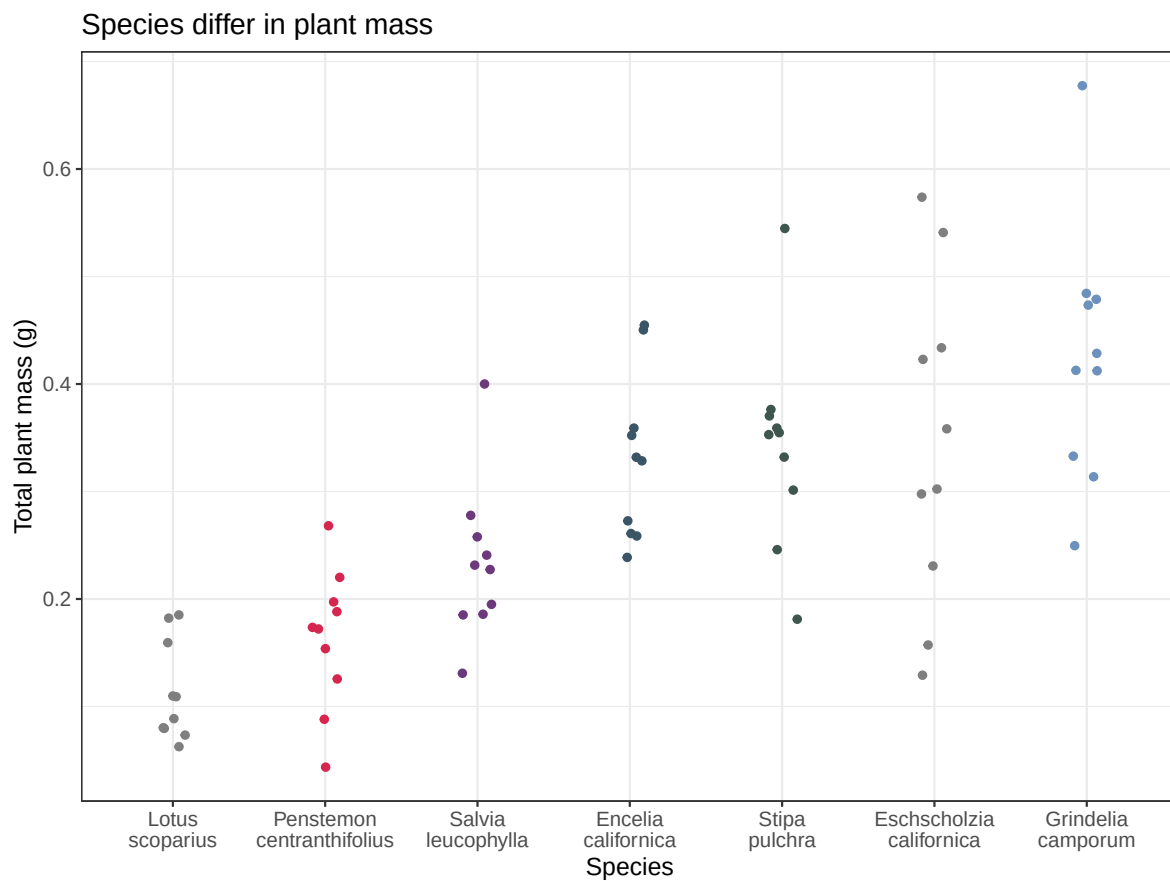
```
ggplot(data = drought_exp_clean,
  mapping = aes(
    x = species_name,
    y = total_g,
    color = species_name)) +
  geom_jitter(
```

```

    height = 0,
    width = 0.1,
  ) +
  scale_x_discrete(labels = label_wrap(10)) +
  labs(title = "Species differ in plant mass",
       x = "Species",
       y = "Total plant mass (g)") +
  scale_color_manual(
    values = c(
      "Lotus scopariis" = lotsco_col,
      "Penstemon centranthifolius" = pencen_col,
      "Salvia leucophylla" = sallev_col,
      "Encelia californica" = enccal_col,
      "Stipa pulchra" = stipul_col,
      "Eschscholzia californica" = esccal_col,
      "Grindelia camporum" = gricam_col
    )
  ) +
  theme(legend.position = "none")

```





#### 4. Fitting models

8 models total:

| Model number | SLA | Water treatment | Species | Predictor list              |
|--------------|-----|-----------------|---------|-----------------------------|
| 0            |     |                 |         | no predictors (null model)  |
| 1            | X   | X               | X       | all predictors (full model) |
| 2            | X   | X               |         | SLA and water treatment     |
| 3            | X   |                 | X       | SLA and species             |
| 4            |     | X               | X       | water treatment and species |
| 5            | X   |                 |         | SLA                         |
| 6            |     | X               |         | water treatment             |
| 7            |     |                 | X       | species                     |

## Model fitting

```
# model 0: null model

model0 <- lm(
  total_g ~ 1,
  data = drought_exp_clean
)

# model 1: all predictors

model1 <- lm(
  total_g ~ sla + water_treatment + species_name,
  data = drought_exp_clean,
)

# model 2: SLA and water treatment

model2 <- lm(
  total_g ~ sla + water_treatment,
  data = drought_exp_clean,
)

# model 3: SLA and species

model3 <- lm(
  total_g ~ sla + species_name,
  data = drought_exp_clean,
)

# model 4: water treatment and species

model4 <- lm(
  total_g ~ water_treatment + species_name,
  data = drought_exp_clean,
)

# model 5: SLA

model5 <- lm(
  total_g ~ sla,
```

```
    data = drought_exp_clean,
  )

# model 6: water treatment

model6 <- lm(
  total_g ~ water_treatment,
  data = drought_exp_clean,
)

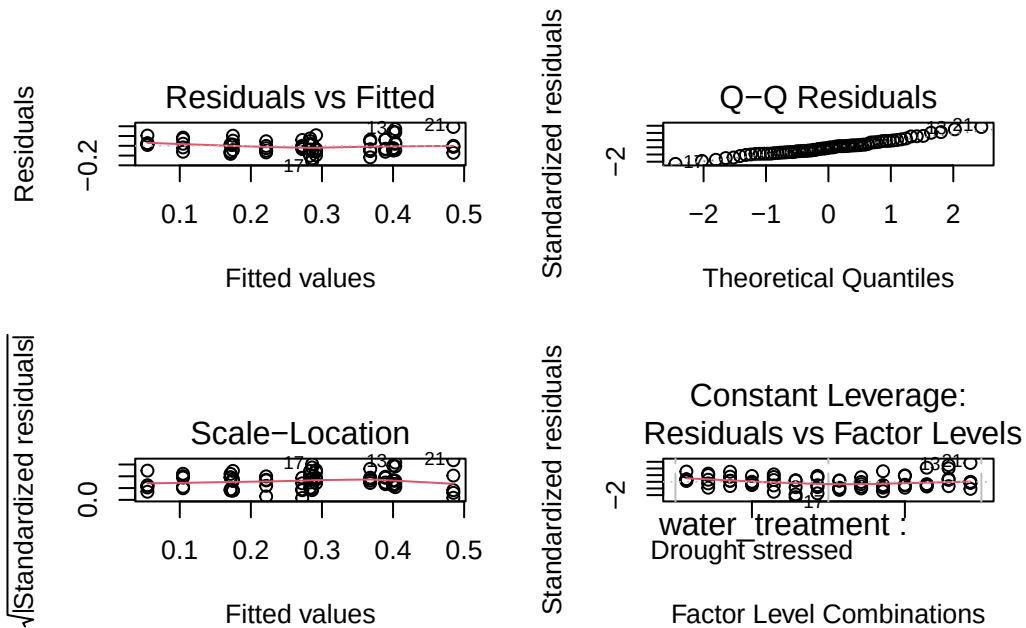
# model 7: species

model7 <- lm(
  total_g ~ species_name,
  data = drought_exp_clean,
)
```

### Model diagnostics (display these after selection)

```
# displaying plots in 2x2 grid

par(mfrow = c(2,2))
plot(model4)
```



- Residuals are normally distributed (QQ plot)
- Residuals are homoscedastic (Residual v. Fitted/Scale Location),
- No outliers that could be influencing model coefficient estimates (Leverage)

## Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
AICc(
  model0,
  model1,
  model2,
  model3,
  model4,
  model5,
  model6,
  model7
) |>
  arrange(AICc)
```

|         | df | AICc       |
|---------|----|------------|
| model4  | 9  | -156.19595 |
| model11 | 10 | -153.75361 |
| model3  | 9  | -124.07569 |
| model7  | 8  | -120.30191 |
| model2  | 4  | -95.82521  |
| model5  | 3  | -88.85180  |
| model6  | 3  | -86.76661  |
| model0  | 2  | -74.98036  |

Insert code to display the summary for the best model.

```
#| label: model-summary
summary(model4)
```

Call:

```
lm(formula = total_g ~ water_treatment + species_name, data = drought_exp_clean)
```

Residuals:

| Min       | 1Q        | Median    | 3Q       | Max      |
|-----------|-----------|-----------|----------|----------|
| -0.157087 | -0.046953 | -0.003733 | 0.041244 | 0.192657 |

Coefficients:

|                                        | Estimate | Std. Error | t value | Pr(> t )    |
|----------------------------------------|----------|------------|---------|-------------|
| (Intercept)                            | 0.05455  | 0.02451    | 2.225   | 0.02973 *   |
| water_treatmentWell watered            | 0.11695  | 0.01733    | 6.746   | 5.90e-      |
| 09 ***                                 |          |            |         |             |
| species_namePenstemon centranthifolius | 0.05003  | 0.03243    | 1.543   | 0.12799     |
| species_nameSalvia leucophylla         | 0.12020  | 0.03243    | 3.706   | 0.00045 *** |
| species_nameEncelia californica        | 0.21774  | 0.03243    | 6.714   | 6.70e-      |
| 09 ***                                 |          |            |         |             |
| species_nameStipa pulchra              | 0.22881  | 0.03243    | 7.055   | 1.72e-      |
| 09 ***                                 |          |            |         |             |
| species_nameEschscholzia californica   | 0.23164  | 0.03243    | 7.143   | 1.22e-      |
| 09 ***                                 |          |            |         |             |
| species_nameGrindelia camporum         | 0.31335  | 0.03243    | 9.662   | 5.53e-      |
| 14 ***                                 |          |            |         |             |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07252 on 62 degrees of freedom  
Multiple R-squared: 0.7535, Adjusted R-squared: 0.7257  
F-statistic: 27.08 on 7 and 62 DF, p-value: < 2.2e-16

### Stop and think: what does this model mean?

#### Best model?

The best model includes water treatment and species as predictors

#### How much variation in the response (total mass, in grams) does this model explain?

About 73%

#### How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

- Species influences plant mass.
- Treatment influences plant mass.
- (Intercept): The intercept is the model prediction for plant mass for *Lotus scoparius* in the *drought stressed* treatment.
- `water_treatmentWell watered`: model prediction for the difference in *Lotus scoparius* mass between well watered and drought stressed treatments.
- `species_namePenstemon centranthifolius`: difference between *Penstemon mass* in the *well watered* treatment and *Lotus scoparius mass* in the *well watered* treatment.
- `species_nameEschscholzia californica`: difference between *Eschsholzia californica mass* in the *well watered* treatment and *Lotus scoparius mass* in the *well watered* treatment.

## 4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

Look at the object before plotting.

```

preds <- ggpredict(
  model4, # model object
  terms = c("water_treatment", "species_name") # name predictors to create a df
) |>
# renaming columns to fit original df
rename(water_treatment = x,
       species_name = group)

```

## 5. Final figure

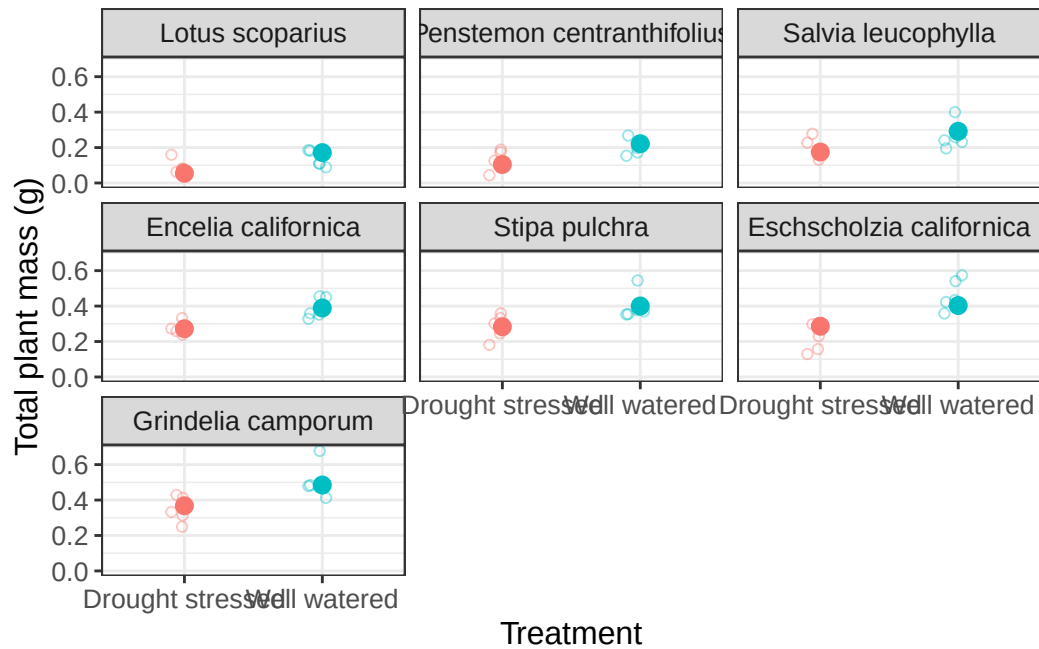
Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```

#| label: final-figure

# base layer: ggplot
ggplot(data = drought_exp_clean,
       mapping = aes(x = water_treatment,
                     y = total_g,
                     color = water_treatment)) +
# jitter: individual plants for each treatment in each species
geom_jitter(height = 0,
            width = 0.1,
            alpha = 0.4,
            shape = 21) +
# pointrange: model prediction with 95% CI
geom_pointrange(data = preds,
               mapping = aes(x = water_treatment,
                             y = predicted,
                             ymin = conf.low,
                             ymax = conf.high)) +
# create different panels for each species
facet_wrap(~ species_name) +
# remove legend
theme(legend.position = "none") +
# relabels x- and y- axes
labs(x = "Treatment",
     y = "Total plant mass (g)")

```



## 6. Writing

### Figure caption

#### Basic components of test

predictors

response

test

distribution

parameters (degrees of freedom)

$R^2$

p-value

significance level

reference level(s)



## **Written communication**

### **Methods**

We used linear models to determine the predictive relationship between specific leaf area ( $\text{cm}^2/\text{g}$ ), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

### **Results**

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model:  $F(2, 67) = 14.57$ ,  $p < 0.001$ ,  $\alpha = 0.05$ ).